

MScFE 610 FINANCIAL ECONOMETRICS

Group Work Project # 1

[See grading rubric here.](#)

Problem 1

Assume that the real regression model for y is the following:

$$y(i) = \alpha + \beta x(i) + \gamma w(i) + \delta z(i) + \varepsilon(i) \quad (1)$$

However, the analyst erroneously estimates the model:

$$y(i) = \alpha + \beta x(i) + \gamma w(i) + \mu(i) \quad (2)$$

Question 1a

If the error terms $\varepsilon(i)$ satisfy the standard assumptions, does it mean that the error terms $\mu(i)$ do as well? Explain why or why not using rigorous formulae.

Question 1b

Assume that we estimate model (2) and that x and w are orthogonal. What are the estimated $\hat{\beta}$ and $\hat{\gamma}$?

Question 1c

Determine explicitly the bias for $\hat{\gamma}$ as estimated from model (2) and using the true model (1) for y . [Note: x and w are assumed orthogonal. Nothing is said about the relationships of x and w with z .]

Question 1d

In what particular case would the estimates of the parameter of the model α , β , and δ obtained from model (2) be the same as those obtained from model (1)?

Question 1e

Using simulations, create a known regression model containing two predictors. In this case the parameters of the model are known. Now omit one of the predictors and estimate the model. Compare the estimates of the coefficient of the non omitted predictor in both circumstances. What do you notice? If you increase the sample size of the simulation, does the result change? (**Hint:** You create a model like

$$Y(i) = a + bX(i) + cZ(i) + e(i)$$

where a , b , c are known. Try simulating X and Z so that they are not necessarily independent and then estimate the model with 2 and only 1 predictor and pay attention to the estimates for b without omitting Z and omitting Z . e should be independent of X and Z , however!!)

Problem 2

Question 2a

Address the sensitivity to outliers for regression models in general. About half a page to a page in length, no need for more!

Question 2b

Then, use a simulated instance to discuss/illustrate how the presence of outliers can change the estimated parameters for a regression model. Make a special effort to describe clearly how your simulation project was put together.

Question 2c

Suppose that you have a dataset consisting of n observations $(x_1, y_1), \dots, (x_n, y_n)$. You are fitting a simple linear regression model:

$$Y_t = \alpha + \beta x_t + \varepsilon_t$$

Imagine that $n-1$ points are clustered tightly along the line $y=0$, that is $(x_i, y_i) = (x_i, 0)$ for $i=1, 2, \dots, n-1$. Now, introduce a single outlier at (x_n, y_n) where x_n is very large (a high leverage point).

Prove that as $x_n \rightarrow \infty$, the OLS estimate for the slope is driven entirely by the ratio y_n/x_n regardless of the other $n-1$ points.

Problem 3

Question 3a

Use one dataset below depending on your Group number:

- Odd Group number: [FE-GWP1_model_seletion_1.csv](#)
- Even Group number: [FE-GWP1_model_selection_2.csv](#)

With the assigned dataset, perform model selection (i.e. find the best regression model for y in terms of x_1, x_2, x_3, x_4 , and x_5 , using a few approaches (at least 2).

Valid approaches consist of (forward or backward selection, using adjusted R-squared, the AIC criterion, the BIC criterion).

It is important that you explain your work. No credit will be given if your model selection appears to be missing any justification and/or is done by a simple trial and error procedure.

Question 3b

Which criterion do you think is more reliable in general? BIC or AIC? This is not an open question! There is a proof as for which one is better. Imagine that a client or a colleague

asked you the same question. HINT: one way to think about this one is to consider two models:

The true model:

$$y(i) = \alpha + \beta x(i) + \varepsilon(i)$$

but we estimate instead a model with an extra and not required predictor, z:

$$y(i) = \alpha + \beta x(i) + \gamma z(i) + \mu(i)$$

A very good selection criterion would pick the true model at least when the sample size n increases to infinity with probability 1.

Problem 4

Time to show our skills in interpreting a regression model.

We want to estimate the determinants of the hourly wage. The dependent variable is typically the natural logarithm of the wage, $\ln(\text{wage})$. The model we use is:

$$\ln(\text{wage}) = \beta_0 + \beta_1 \text{educ.} + \beta_2 \text{gender} + \beta_3 \text{age} + \beta_4 \text{age}^2 + \beta_5 \text{degree} + \varepsilon.$$

where:

- **Educ.:** Years of schooling.
- **Gender:** Dummy variable (1 if Female, 0 if Male).
- **Age:** Age in years.
- **Degree:** Dummy variable (1 if they hold a Master's degree or higher, 0 otherwise).

Question 4a

Why are we using age^2 in the regression? (a 4-5 line answer is sufficient).

Question 4b

Why are we using $\ln(\text{wage})$ instead of wage as the dependent variable (4-5 line answer is sufficient).

After running an OLS regression on a sample of 2,000 working professionals, we obtained the following estimates:

Variable	Coefficient	Std. Error	t-stat
<i>intercept</i>	0.85	0.12	7.08
<i>educ.</i>	0.08	0.01	8.00
<i>gender</i>	-0.15	0.10	-1.50
<i>age</i>	0.19	0.06	3.17
<i>age^2</i>	-0.002	0.0003	-6.67
<i>degree</i>	0.22	0.04	5.50

Model statistics:

R-squared: 0.44

Sample size (n): 2000

Question 4c

Why do we use $\ln(\text{wage})$ instead of wage? It is a simple question, nothing to do with mathematical reasons.

Question 4d

What is the interpretation of $\beta_1 = 0.08$?

Question 4e

Is there a gender gap?

Question 4f

Based on the estimated model, at what age is salary per hour the highest?

Question 4g

What is the average impact of holding an advanced degree on hourly salary?

Question 4h

Could we have used two dummy variables, say “male” and “female” instead of just “gender”?

Problem 5

It is now time to show your worth dealing with nonstationary time series and regressions.

We are given 3 observed time series X, Y, and Z whose underlying models are:

$$X(t) = 1.0w(t) + \eta(t)$$

$$Y(t) = 0.5w(t) + \theta(t)$$

$$Z(t) = -1.0w(t) + \chi(t)$$

where

$w(t) = w(t-1) + \varepsilon(t)$, and the terms η, θ, χ are $I(0)$ white noise components.

Question 5a

Assume that $w(0) = 0$. Prove that

$$X(t) = \sum_{i=0}^t \varepsilon(i) + \eta(t).$$

Is X a stationary ($I(0)$) process? How about Y and Z?

Question 5b

Can you tell how many cointegration vectors are there for our variables X, Y, and Z? Identify the cointegrated vectors.

Question 5c

Using the simulated data contained in [dataset_Prob5.csv](#):

1. Perform a visual inspection of the time series
2. Perform unit root testing for each time series

Complete cointegration testing using the Johansen test (trace and maximum eigenvalue statistics) and report the appropriate cointegration rank

Finally, estimate the VECM for X, Y and Z.

Problem 6

Suppose you have an AR(1) process where $Y(t) = 0.5Y(t-1) + \varepsilon(t)$

Question 6a

Write out the first four terms of its MA(∞) representation. [NOTE: you must show your steps, copying the answer from a textbook or using AI to derive it without details will provide no credit].

Assume now that the process Z follows an AR(2) process;

$$Z(t) = 0.6Z(t-1) + 0.2Z(t-2) + \varepsilon(t).$$

Question 6b

Find the autocorrelation of order 1 and 2 for the process Z.

Groups

Groups of 3 students need to answer the questions for all 6 Problems.

Groups of 2 students are required to answer the questions for Problems 1, 2, 4, 5, and 6.

Groups of 1 student need to address the questions for Problems 1, 2, 3, and 5.

Submission Requirements and Format

- 1) ****1 PDF document with the answer to all questions.** This PDF **must include all the comments, plots, tables, and the required results.** Do not send the readers back and forth to the Notebook. You are writing for readers/clients.

Be sure to include the specific question (e.g. 2a, 3b, etc.) before you put your answer. That means your submission will include every question number before the respective answer.

The code used should only appear in the notebook, see below.

- a) Use the available Report Template and fill out the required information in the first page.
- 2) A **zipped folder** including:

- a) An executable Jupyter notebook (.ipynb) that addresses all the challenges
- b) A duplicate version of the Jupyter notebook code and output in PDF or HTML format.
 - i) In order to include the output of the code, you must RUN the code before downloading the PDF.

One team member submits the above on behalf of the entire group.

If you miss the proper submission, you will be penalized 10 points. At this point in your career as students at WQU you should know the drill. No excuses. Even if you are allowed to resubmit, the 10 point penalization will not be removed. Therefore PAY ATTENTION!

Direct copy/paste from AI tools and any use of AI tools are NOT ALLOWED. Heavy penalizations would be assessed if use of AI is detected. The purpose of the GWP is not to assess how well ChatGPT or other AI tools can answer the questions, but how the students would. **Your assignment will be checked against AI to determine if it was used.**

READ the [Academic Policy on the Use of AI](#): in-text citations and references are **mandatory**, use [this link](#) to learn how to add them.

Make sure that all paper has a uniform format. This is a group task.. If each student uses different formats, it means that it was not really done as a group. Penalizations will be assessed if there are different fonts, sizes for different questions as this is evidence of the lack of group collaboration.

Rubric

Your instructor will evaluate your group submission for GWP1 using the following rubric:

Quantitative Analysis (Open-Ended Questions)	Technical and Non-Technical Reports	Writing and Formatting
60 Points	40 Points	20 Points
The group is able to apply results, formulas, and their knowledge of theory to real-life finance scenarios by doing the following: • Providing all the necessary information to support their arguments. • Presenting arguments that reflect group discussion and research. • Using authoritative references to support a position and provide updated information. • Concluding with practical takeaways for more insightful financial decision-making.	Technical Reports contain 3 parts: 1) code for each question (be sure to explicitly state the question number), 2) the corresponding output of that code, and 3) <u>interpretations and/or recommended courses of action that reasonably follow from those results.</u> Note: Technical reports will include the technicalities of models, such as names, methods of estimation, parameter values, etc., and exclude generalities about the work done. It should NOT include names of Python code that were used.	A submission that looks professional should: • Include the axes, labels, and scales in graphs. • Be free of significant grammatical errors or typos. • Be an organized, well-structured, and easy-to-read document. • Include proper citations and a bibliography in MLA format.
	Non-Technical Reports contain 3 parts: 1) clear explanation of results; 2) the recommended course of action that follows; and 3) the identification of factors that impact each portfolio. Note: AVOID all references to model names, algorithms, and unnecessary details. Instead, focus on the investment decision.	

Revision Date: May 7, 2025